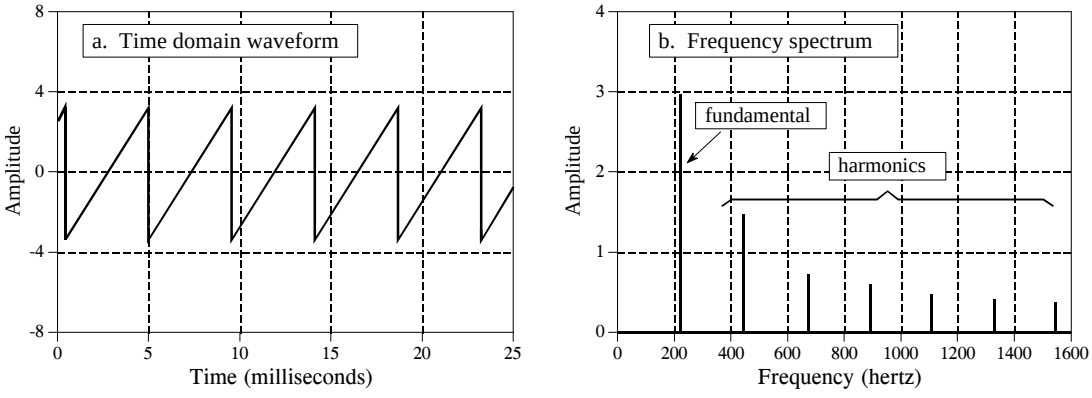

FIGURE 22-3
Violin waveform. A bowed violin produces a sawtooth waveform, as illustrated in (a). The sound *heard by the ear* is shown in (b), the fundamental frequency plus harmonics.

This is the basis of the standard musical scale, as illustrated by the piano keyboard in Fig. 22-4. Striking the farthest left key on the piano produces a fundamental frequency of 27.5 hertz, plus harmonics at 55, 110, 220, 440, 880 hertz, etc. (there are also harmonics between these frequencies, but they aren't important for this discussion). These harmonics correspond to the fundamental frequency produced by other keys on the keyboard. Specifically, every *seventh* white key is a harmonic of the far left key. That is, the eighth key from the left has a fundamental frequency of 55 hertz, the 15th key has a fundamental frequency of 110 hertz, etc. Being harmonics of each other, these keys sound similar when played, and are harmonious when played in unison. For this reason, they are *all* called the note, *A*. In this same manner, the white key immediate right of each *A* is called a *B*, and *they* are all harmonics of each other. This pattern repeats for the seven notes: *A*, *B*, *C*, *D*, *E*, *F*, and *G*.

The term **octave** means a *factor of two in frequency*. On the piano, one octave comprises eight white keys, accounting for the name (*octo* is Latin for *eight*). In other words, the piano's frequency doubles after every seven white keys, and the entire keyboard spans a little over seven octaves. The range of human hearing is generally quoted as 20 hertz to 20 kHz,

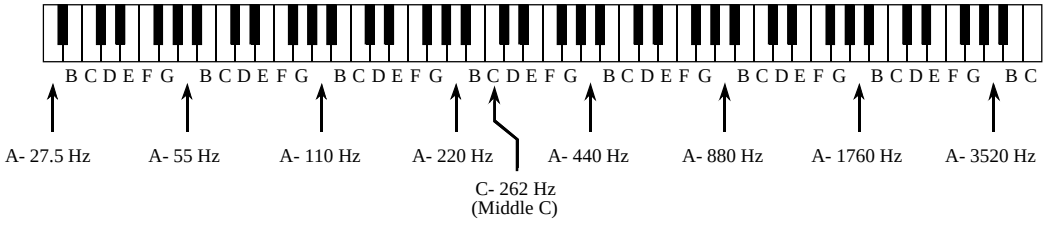

FIGURE 22-4
The Piano keyboard. The keyboard of the piano is a *logarithmic* frequency scale, with the fundamental frequency doubling after every seven white keys. These white keys are the notes: *A, B, C, D, E, F and G*.